

Wang Xinyi

Address: 17W, Hong Kong Science Park, Shatin, N.T., Hong Kong.

E-mail: xywangmae@link.cuhk.edu.hk

Tel: (+852) 53921596

Education

- **Postdoctoral Fellow**, Hong Kong Centre For Logistics Robotics, Department of Mechanical & Automation Engineering, The Chinese University of Hong Kong 10/2023-Now
- **Ph.D. Degree**, Department of Mechanical & Automation Engineering, The Chinese University of Hong Kong 08/2019-09/2023
Supervisor & Co-supervisor: Prof. Ben M. Chen (IEEE Fellow & Academy of Engineering Fellow, Singapore) & Prof. Jie Huang (IFAC Fellow & IEEE Fellow)
- **B.Eng. Degree**, Flight Vehicle Design and Engineering, Xiamen University 09/2015-06/2019
GPA: 91.8/100 (Ranking 1st out of 46)
- **Visiting student**, University of Oxford 02/2017
- **IELTS:** 6.5 (R6.5, L6.5, S6.0, W6.0) 02/2018

Research Interests

Task and motion planning, multi-agent coordination system, optimization and control theory

Publications

● Journals & conference

- [1] **X. Wang**, J. Xu, C. Gao, Y. Chen, J. Zhang, C. Wang and B. M. Chen, Sensor-based multi-agent coverage control with spatial separation in unstructured environments. 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- [2] **X. Wang**, Y. Ding, Y. Chen, R. Han, L. Xi, B. M. Chen. OA-ECBVC: A Cooperative collision-free encirclement and capture approach in cluttered Environments. IEEE Conference on Decision and Control (CDC), 2023.
- [3] Y. Chen, R. Wang, **X. Wang**, B. M. Chen. Sampling-based path planning under temporal logic constraints with real-time adaptation. IEEE International Conference on Robotics and Automation (ICRA), 2023.
- [4] **X. Wang**, L. Xi, Y. Ding, B. M. Chen. Distributed encirclement and capture of multiple pursuers with collision avoidance. IEEE Transactions on Industrial Electronics (TIE), 2023.
- [5] Y. Chen, **X. Wang**, R. Wang, Z. Guo, S. Lai, B. M. Chen. An interactive system for multiple-task linear temporal logic path planning. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
- [6] **X. Wang**, C. Gao, X. Chen, B. M. Chen. Fast and secure distributed multi-agent coverage control with an application to infrastructure inspection and reconstruction. Chinese Control Conference (CCC), 2023, Guan Zhao-Zhi Award Finalist.
- [7] C. Gao, **X. Wang**, X. Chen, B. M. Chen. A hierarchical multi-UAV cooperative framework for infrastructure inspection and reconstruction. Control Theory and Technology (CTT), 2023.
- [8] **X. Wang**, L. Xi, Y. Chen, S. Lai, F. Lin, B. M. Chen. Decentralized MPC-based trajectory generation for multiple quadrotors in cluttered environments. Guidance, Navigation and Control (GNC), 2021, 1(02): 2150007.

● Patents

- [1] B. M. Chen, Y. Chen, R. Wang, **X. Wang**, Xi Chen. A Human-in-the-Loop System for Mobile Robot Task Planning. Patent of the United States, No. 63/450,314, Mar. 6, 2023.

Technical Skills

- Experienced in C/C++, Python, MATLAB, LaTeX, Git, ROS, Gazebo

Courses Experience

- ENGG 5501 Foundations of Optimization, instructed by Professor Anthony Man-Cho So.
- MAEG5403 Linear System Theory & Design by Professor Ben M. Chen
- MAEG5070 Nonlinear Control Systems by Professor Jie Huang

Awards & Honors

Postdoctoral Scholarship, the Innovation and Technology Fund of Hong Kong	10/2023-Now
Professor Charles K. Ko Student Creativity Award, CUHK	06/2023
Research Postgraduate Studentship, CUHK	2019-2023
Outstanding Graduates, XMU (Top 2%)	06/2019
Merit Student, XMU (Twice, Top 8%)	1 2/2016&12/2017
Xiamen International Bank Scholarship, XMU (Top 4%)	04/2017

Academic Services

● **Reviewer of Journals**

- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Cybernetics
- International Journal of Robust and Nonlinear Control
- Guidance, Navigation and Control

● **Reviewer of Conferences**

- Associate Editor of IEEE International Conference on Control and Automation (ICCA)
- IEEE Conference on Decision and Control (CDC)
- Chinese Control Conference (CCC)

● **Presentations**

- ICRA 2023, London, England, poster presentation
- CDC 2023, Singapore, oral presentation
- CCC 2023, Tianjing, China, present for Guan Zhao-Zhi Award
- ICRA 2024, Yokohama, Japan, oral presentation